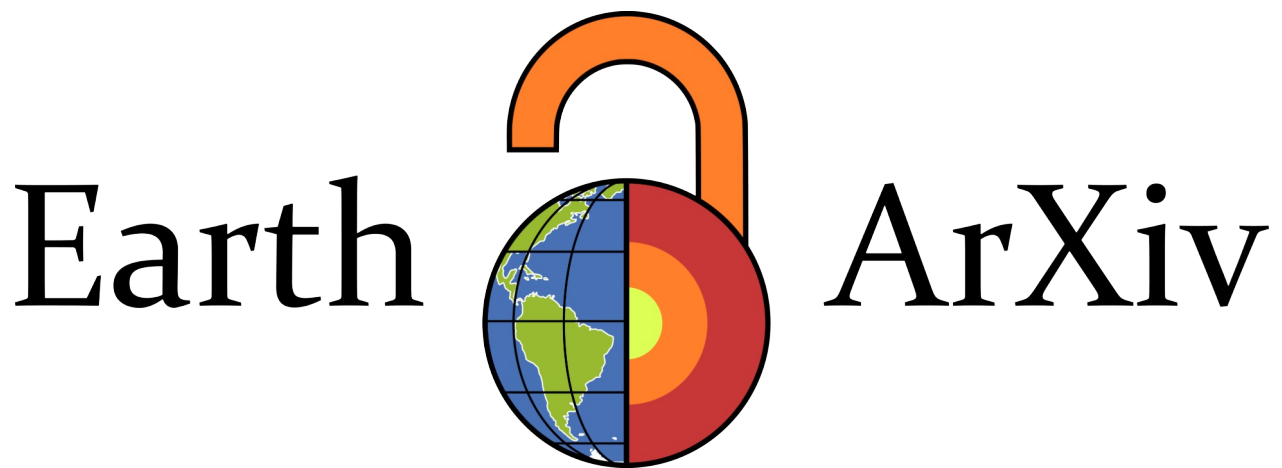




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Prepared in cooperation with Kankakee River Basin and Yellow River Basin
Development Commission

Evaluating Bank Restoration and Stabilization in the Yellow River in Northern Indiana: Mitigation of Sediment and Nutrient Transport

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was not included for sediment and nutrient concentrations and loads in the Yellow River remediation study in northwest Indiana.

Conversion Factors

U.S. customary units to International System of Units

International System of Units to U.S. customary units

Multiply	By	To obtain
Length		
kilometer (km)	0.6214	mile (mi)
meter (m)	1.094	yard (yd)
Area		
square meter (m ²)	0.0002471	acre
hectare (ha)	2.471	acre
square hectometer (hm ²)	2.471	acre
square kilometer (km ²)	247.1	acre
hectare (ha)	0.003861	square mile (mi ²)
square kilometer (km ²)	0.3861	square mile (mi ²)
Volume		
cubic centimeter (cm ³)	0.06102	cubic inch (in ³)
cubic decimeter (dm ³)	61.02	cubic inch (in ³)
liter (L)	61.02	cubic inch (in ³)
cubic decimeter (dm ³)	0.03531	cubic foot (ft ³)
cubic meter (m ³)	35.31	cubic foot (ft ³)
Flow rate		
meter per second (m/s)	3.281	foot per second (ft/s)
meter per hour (m/h)	39.37	inch per hour (in/h)
millimeter per year (mm/yr)	0.03937	inch per year (in/yr)
kilometer per hour (km/h)	0.6214	mile per hour (mi/h)
Mass		
gram (g)	0.03527	ounce, avoirdupois (oz)
kilogram (kg)	2.205	pound avoirdupois (lb)
metric ton (t)	1.102	ton, short [2,000 lb]
metric ton (t)	0.9842	ton, long [2,240 lb]

Temperature in degrees Celsius (°C) may be converted to degrees Fahrenheit (°F) as follows:

$$^{\circ}\text{F} = (1.8 \times ^{\circ}\text{C}) + 32.$$

Temperature in degrees Fahrenheit (°F) may be converted to degrees Celsius (°C) as follows:

$$^{\circ}\text{C} = (^{\circ}\text{F} - 32) / 1.8.$$

Datums

Vertical coordinate information is referenced to the North American Vertical Datum of 1988 (NAVD 88).

Horizontal coordinate information is referenced to the North American Datum of 1983 (NAD 83).

Supplemental Information

Turbidity is in Formazin Nephelometric Units (FNU).

Abbreviations

Basin commission Kankakee River Basin Yellow River Basin Development Commission

USGS U.S. Geological Survey

NIST National Institute of Standards and Technology

LRT Likelihood ratio test

GLS generalized least squares

LMM linear mixed models

Abstract

The U.S. Geological Survey, in cooperation with the Kankakee River Basin and Yellow River Basin Development Commission, evaluated the effects of streambank remediation on the Yellow River in northwest Indiana. Two phases of remediation were completed on the Yellow River between 2021 and 2023, after assessments indicated it was a large contributor of sediment to the Kankakee River. Temporal effects were assessed using pre-remediation (April 2019 – October 2021) and post-remediation (February – October 2024) data from two long-term monitoring sites, USGS 05516665 Yellow River near Oak Grove (Oak Grove) and USGS 05517000 Yellow River at Knox (Knox). Spatial patterns were evaluated among sites from upstream to downstream during the post-remediation period. Monitoring included the pre-existing Oak Grove and Knox sites, an additional monitoring site at USGS streamgage 05516500 Yellow River at Plymouth (Plymouth), and discrete sampling at two additional sites. Discrete sites include USGS 05516663 Yellow River at Upas Rd near Oak Grove (Upas Rd), located upstream of the remediation reach, and USGS 05516667 Yellow River at CR11100E near Oak Grove (CR1100E), located within the remediation reach.

Generalized least squares and linear mixed-effects regression models were used to evaluate temporal differences in turbidity-to-discharge relations for slope and turbidity units, discharge, and suspended-sediment in before and after remediation data. Similar analysis using generalized least squares and linear mixed-effects regression models were conducted for spatial analysis of discharge and sediment and nutrient concentrations and fluxes at each site post-remediation. Temporal comparisons showed that turbidity increased less with discharge increases after remediation, indicated by slope of relation, at both Oak Grove and Knox,

indicating improved water clarity during higher flows. Mean turbidity and suspended-sediment concentrations were not statistically different between remediation periods.

Comparison of the post-remediation continuous water quality data from upstream to downstream sites through the remediated reaches indicated that downstream sites exhibited steeper turbidity-to-discharge slopes than the upstream control site at Plymouth. A statistically significant difference was detected between the slopes at Oak Grove and Plymouth, whereas Knox did not differ significantly from either Plymouth or Oak Grove. Downstream sites had significantly lower mean turbidity, suggesting improved water clarity within and downstream of the remediation reach. Nutrient concentrations generally decreased moving downstream and suspended-sediment concentrations showed no consistent trend. Nutrient and sediment fluxes increased downstream, likely reflecting the influence of greater discharge volumes.

Interpretation of remediation effects was limited by the absence of pre-remediation data at upstream sites and by the short post-remediation monitoring period. Continued monitoring could allow for assessment of longer-term responses to streambank remediation on the Yellow River, as effects may become more pronounced over time.

Plain Language Summary

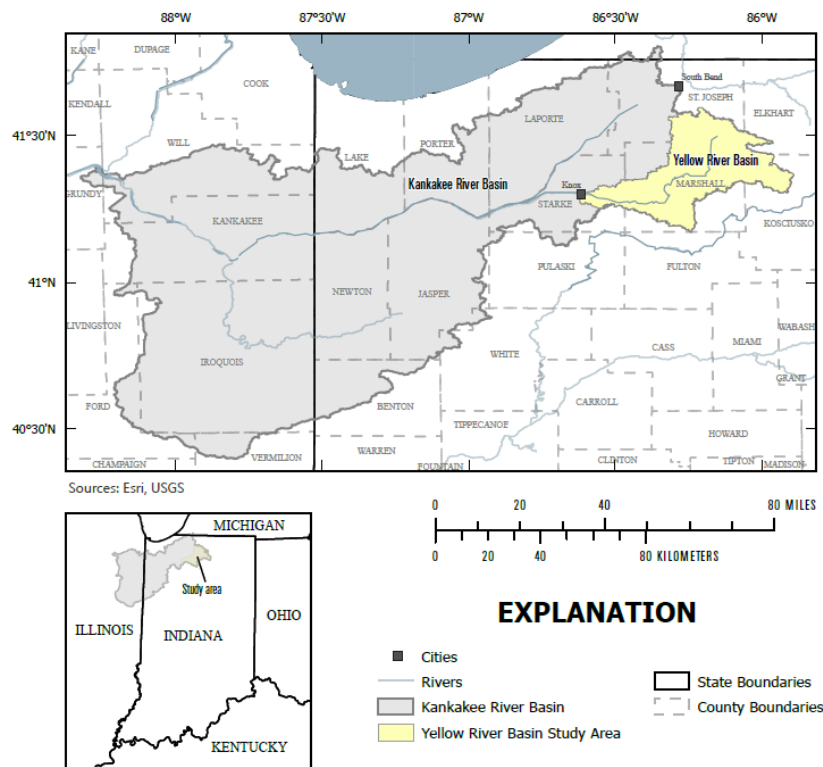
Human activity has substantially shaped the Kankakee River Basin in Indiana, including widespread settlement and agriculture near the Yellow and Kankakee Rivers. These modifications have led to increased sediment and nutrient inputs into waterbodies, contributing to downstream sedimentation and excess nutrients. In 2024, the U.S. Geological Survey (USGS) expanded monitoring and sampling on the Yellow River to evaluate how streambank remediation may be affecting sediment and nutrient transport within the river. This study examined pre- and

post-remediation conditions at two sites and examined upstream-to-downstream patterns across five sites, following two remediation phases completed from 2021 to 2023. At sites where data is available for before and after remediation efforts, there was evidence of improved water clarity, shown by generally lower relations in discharge and turbidity after remediation. Post-remediation data collected along the river in 2024 indicated nutrient concentrations generally decreased moving from upstream to downstream sites, whereas sediment concentrations showed no consistent increase or decrease downstream. Nutrient and sediment fluxes increased from upstream to downstream sites, but this was likely due to higher discharge, rather than increases in concentration. Overall, remediation sites showed some signs of reduced concentrations, but downstream hydrology continued to drive the amount of material carried downstream. As remediation efforts become more established and expand in the watershed, these improvements in water quality may become more pronounced.

Introduction

The greater Kankakee River Basin extends from northwest Indiana to northeast Illinois (fig. 1). The basin once included the Grand Kankakee Marsh, historically the largest wetland in the Midwestern United States (Meyer, 1935). In the late 19th and early 20th century, the Grand Kankakee Marsh was drained by channelization of the rivers; a process of deepening, widening, and straightening the rivers (Little Jr and Jonas, 2013; Demissie and others, 2016). Additionally, extensive sub-surface drainage (tile systems) was implemented to more efficiently remove excess water from the land so that the fertile soil surrounding the river could be used for agriculture (Mitsch and Gosselink, 2000; Little Jr and Jonas, 2013; Demissie and others, 2016). This process resulted in the elimination of most of the wetlands in the Kankakee River

Basin (Mitsch and Gosselink, 2000). Increased bank slopes and water velocities caused by channelization in the basin have been linked to increased erosion and sediment transport (Little Jr and Jonas, 2013). As a result, stakeholders including The Kankakee River Basin and Yellow River Basin Development Commission (hereafter referred to as “the Basin Commission”) have undergone efforts to identify and mitigate erosion and sediment transport within the basins.



The Yellow River is a major source of sandy sediment to the Kankakee River because its steep banks and sandy channel make it highly susceptible to erosion (Little Jr. and Jonas, 2013). Located upstream of the Kankakee River Basin, the HUC 10 (0712000105) Headwaters Yellow River Watershed is listed on Indiana Department of Environmental Management's 303(d)

impaired waters list for impaired biotic communities, elevated *Escherichia coli*, and high nutrient levels (Indiana Department of Environmental Management, 2017). Since these headwaters strongly influence downstream conditions, including the focus area of this study, comprehensive sediment and nutrient monitoring and management are critical to protect ecological resources and public health in the Yellow River, Kankakee River, and further downstream reaches (Kelly and others, 2019).

To help inform management decisions in the greater Kankakee River Basin, the U.S. Geological Survey (USGS) in cooperation with the Basin Commission installed a continuous water-quality monitoring network along the Yellow and Kankakee Rivers. In 2016, the network consisted of three monitoring sites (hereafter referred to as sites) on the Kankakee River, one site on the Iroquois River, and three sites on the Yellow River (table 1, described below). Sites on the Kankakee and Iroquois Rivers continuously monitored river gage height, discharge, water temperature, pH, dissolved oxygen, specific conductance, turbidity, and nitrate plus nitrite while the sites on the Yellow River monitored river gage height, discharge, temperature, and turbidity. Discrete water quality samples were also collected and analyzed for suspended sediment at all sites, and for nitrogen and phosphorus constituents at the Kankakee and Iroquois River sites. Before this study, the relation between discrete and continuous data was used to estimate continuous concentrations and fluxes of suspended sediment at the Yellow River and Kankakee River sites (Lathrop and others, 2019; Downhour and Piotrowski, 2025).

In 2019, basin management recommendations were presented in the “Kankakee River Flood and Sediment Management Work Plan”, which outlines a 40-year schedule of sediment management actions for the Kankakee River Basin (Christopher B. Burke Engineering, LLC, 2019). Management actions on the Yellow River were prioritized, with several scheduled for

implementation during 2020 –2026. Bank reconstruction along sections of the Yellow River was planned in four phases. By the start of data collection for this study in 2024, phases 1 and 2 had been completed, phase 3 was in progress, and phase 4 remained in the design stage. Collectively, these phased efforts are hereafter referred to as remediation. Photos before, during, and after remediation for Phase 1 in parts of the Yellow River are shown in figure 2. Other ongoing mitigation efforts in the Kankakee River Basin include log jam management, bridge removal or replacement, berm maintenance, sediment traps and storage areas for runoff events, and flood-protection measures (Christopher B. Burke Engineering, LLC, 2019).



Figure 2. Photographs taken before, during, and after Phase 1 remediation at locations 30–150 meters upstream from USGS 05516665 Yellow River near Oak Grove, Indiana. Photographs provided by Scott Pelath, Executive Director of the Kankakee River Basin and Yellow River Basin Development Commission. A) Before remediation, December 2020; B) during remediation construction, September 2021; C) after remediation, September 2024; and D) after remediation, November 2024.

Following the completion of the first two phases of bank reconstruction in the Yellow River (2023), the USGS, in cooperation with the Basin Commission, expanded water-quality monitoring along the Yellow River. In addition to the two long-term Yellow River monitoring sites used in this study, a continuous monitoring site was added upstream from the remediated reaches, and two additional discrete-sampling sites were selected near the remediated reaches. Thus, there were three sites within the study area collecting real-time data and five sites where discrete samples were collected for both sediment and nutrient concentrations. The goal of this study was to assess how remediation efforts may have affected sediment and nutrient concentrations and flux dynamics in the Yellow River.

Purpose and Scope

This report describes changes in water quality in the Yellow River and evaluates how those changes may relate to remediation efforts. The study focuses on two overarching questions:

- (1) Did water quality differ between the pre-remediation and post-remediation periods?
- (2) Did water quality differ among sites along the Yellow River after remediation?

Data from selected Yellow River sites was used to statistically evaluate temporal differences before (April 2019– August 2021) and after (February 2024– October 2024) remediation, as well as spatial differences from upstream to downstream after completion of phases 1 and 2 of remediation.

Study Area and Site Selection

The Yellow River Basin has a drainage area of 438 mi² and lies within the southeastern portion of the greater Kankakee River Basin. The Yellow River begins south of South Bend,

Indiana and reaches west of Knox, Indiana before the confluence with the Kankakee River (fig. 1). The downstream reaches of the Yellow River are within the Kankakee Outwash and Lacustrine Plain physiographic unit, characterized by broad, flat areas with poor drainage and covered with glacial outwash, dune sand, alluvial deposits and lake sand, while the upstream portion of the Yellow River is within the Steuben Morainal Lake Area physiographic unit, consisting of undulated till plains (Fenelon and others, 1992). Surface deposits along the Yellow River consist of well sorted sand and gravel that were deposited during the retreat of the last continental glaciers (Ivens and others, 1981).

Sites selected for this study (table 1) were chosen based on remediation efforts from Phase 1 (completed by 2021) and Phase 2 (completed 2023) of The Basin Commission's work plan (table 2). This included two long-term water-quality streamgage sites — USGS 05516665 Yellow River near Oak Grove, IN (Oak Grove), and USGS 05517000 Yellow River at Knox, IN (Knox) — where continuous and discrete data were collected before and after remediation. A newly installed water-quality streamgage, USGS 05516500, Yellow River at Plymouth, IN (Plymouth), provided continuous and discrete data after remediation. Discrete water-quality samples also were collected after remediation at USGS 05516663 Yellow River at Upas Rd near Oak Grove, IN (Upas Rd), and USGS 05516667 Yellow River at CR1100E nr Oak Grove, IN (CR1100E) (tables 1 and 2; fig. 3). For clarity, site short names are followed by (U) for sites upstream from remediation, (R) for sites within the remediated reach, and (D) for sites downstream from remediation, as described in table 1.

Land use in the Yellow River Basin study area is dominated by agriculture but varies slightly from the most upstream to the most downstream site (table 3; USGS, 2024). In 2024, agricultural land use in the basin was dominated by cultivated crops which decreased from 77.00

percent at the most upstream site, Plymouth (U), to 70.20 percent at the most downstream site, Knox (D). Developed land accounted for 12.09 to 12.75 percent of the drainage basins and consisted primarily of developed open space and low-intensity development. Table 3 provides a breakdown of land use for the drainage basins of the Plymouth (U), Oak Grove (R), and Knox (D), water-quality streamgages.

1 **Table 1.** Site locations, characteristics and data used in the Yellow River remediation study, northwest Indiana.

[Site information from U.S. Geological Survey (2025); USGS, U.S. Geological Survey; U, upstream from remediation; R, within remediation; D, downstream from remediation; IN, Indiana; SS; suspended sediment; Rd, road; nr, near; n.a., not applicable; ~, approximately; CR 1100E, County Road 1100 East]

Short name	Remediation Status	USGS station name	USGS station number	Continuous Parameters	Link to Continuous Data	Discrete Parameters (2019 –2022)	Discrete Parameters (2024)	Drainage area, in square miles	Remediation Details
Plymouth	U	Yellow River at Plymouth, IN	05516500	Discharge, Temperature, Turbidity	https://waterdata.usgs.gov/monitoring-location/USGS-05516500	n.a.	Nutrients ^a and SS	294	~18 river miles upstream from remediation phase 1
Upas Rd	U	Yellow River at Upas Rd nr Oak Grove, IN	05516663	n.a.	n.a.	n.a.	Nutrients ^a and SS	377	~ 1 river miles upstream from remediation phase 1
Oak Grove	R	Yellow River nr Oak Grove, IN	05516665	Discharge, Temperature, Turbidity	https://waterdata.usgs.gov/monitoring-location/USGS-05516665	SS	Nutrients ^a and SS	377	~0.6 river miles within remediation phase 1
CR1100 E	R	Yellow River at CR 1100E nr Oak Grove, IN	05516667	n.a.	n.a.	n.a.	Nutrients ^a and SS	379	after ~2 river miles of phase I remediation - beginning of phase 2
Knox	D	Yellow River at Knox, IN	05517000	Discharge, Temperature, Turbidity	https://waterdata.usgs.gov/monitoring-location/USGS-05517000	SS	Nutrients ^a and SS	435	~1 river mile downstream from phase 2 of remediation

^a Nutrients include: Total Nitrogen as N, water, unfiltered; Nitrate plus nitrite as N, water, filtered; Total ammonia as N, water, filtered; Nitrite as N, water, filtered; Phosphorus, as P, water, unfiltered; Orthophosphate as P, water, filtered

2

Table 2. Timeline of data collection at study sites and remediation activities on the Yellow River in northwest, Indiana. Site short names are defined in table 1.

Calendar Year	Continuous and Discrete Data Sites	*Discrete Data Only Sites	Continuous Parameters	Discrete Parameters	Remediation Details
2019	Oak Grove & Knox		Discharge and Turbidity	Suspended Sediment Concentration	
2020	Oak Grove & Knox		Discharge and Turbidity	Suspended Sediment Concentration	
2021	Oak Grove & Knox		Discharge and Turbidity	Suspended Sediment Concentration	Phase 1 - about 1.25 consecutive river miles - 09/2021 to 12/2022
2022	Oak Grove & Knox		Discharge and Turbidity	Suspended Sediment Concentration	
2023	Oak Grove & Knox		Discharge and Turbidity	Suspended Sediment Concentration	Phase 2 - about 1.2 non - contiguous river miles - 07/2023 to 12/2023
2024	Plymouth, Oak Grove, & Knox	Upas & 1100 E (Upstream and downstream Oak Grove)	Discharge and Turbidity	Suspended Sediment Concentration and Dissolved and Total Nutrient Concentration	Phase 3 - in progress and not considered for this study

Table 3. Percent land use within the drainage basin of each of the three continuous water-quality sites on the Yellow River in northwest Indiana. Site short name from table 1.

[Data from U.S. Geological Survey, 2024]			
Land use type	Percent land use by site		
	Knox	Oak Grove	Plymouth
Agriculture	70.23	74.76	77.00
Developed	12.60	12.75	12.09
Forest	9.92	6.68	5.92
Wetland	5.57	5.05	4.41
Open water	0.56	0.41	0.36
Other	1.13	0.35	0.22

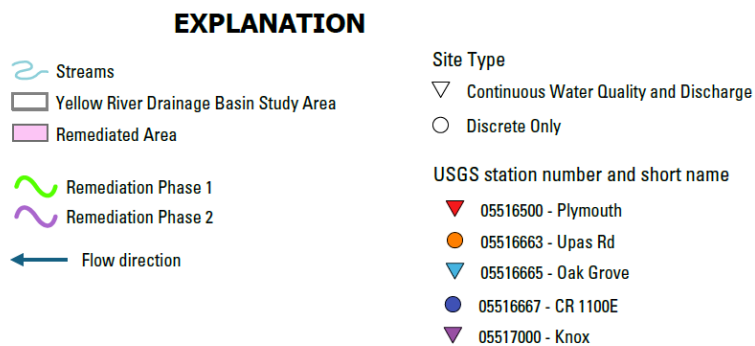
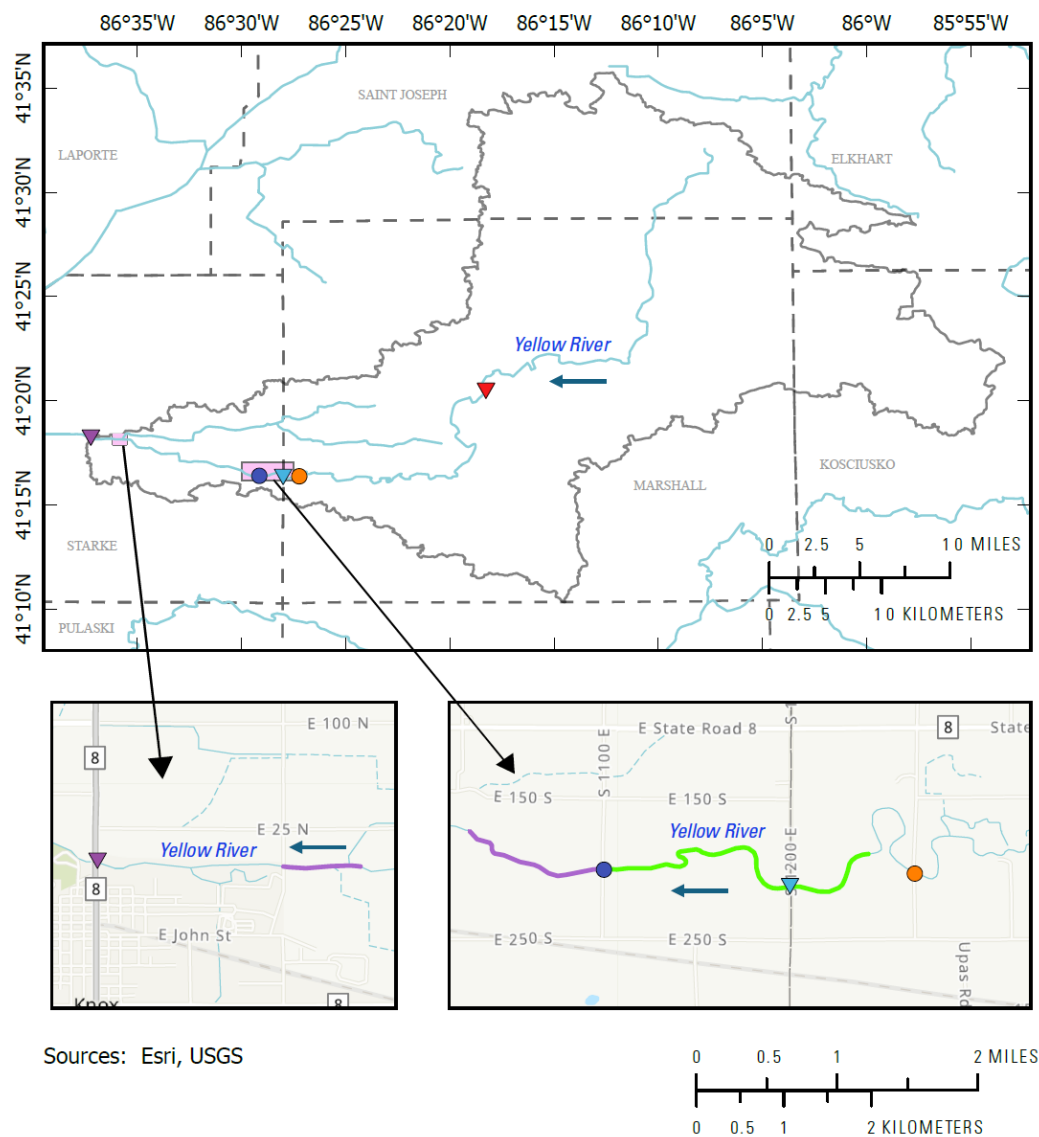


Figure 3. Locations of U.S. Geological Survey (USGS) streamgages, discrete-sampling sites, and remediation extents in the Yellow River study area, northwest Indiana.

Approach and Methods for Data Collection and Analysis

Continuous water-quality and streamflow data used to evaluate remediation efforts on the Yellow River were obtained from USGS streamgages (fig. 3, table 1). Parameters used in the study included gage height (ft), discharge (ft^3/s), and turbidity (FNU). Prior to 2024, continuous water-quality data were available only at the Oak Grove (R) and Knox (D) streamgages. In February 2024, continuous water-quality monitoring was expanded to include the Plymouth (U) site. Discrete water-quality samples were collected at all five monitoring sites in from February to October 2024 and analyzed for suspended-sediment concentration; total nitrogen and total phosphorus; and dissolved nutrient species, including nitrate plus nitrite, nitrate, total ammonia, and orthophosphate. Before 2024, suspended-sediment concentration was measured only at Oak Grove (R) and Knox (D).

Question One, “Did water quality differ before and after remediation?” was addressed by applying statistical tests to compare pre- and post-remediation data. Question Two, “Did water quality differ among sites along the Yellow River after remediation?” was addressed by comparing post-remediation data among sites.

For all analyses in this study, the pre-remediation period was defined as April 2019 through August 2021, and the post-remediation period was defined as February 2024 through October 2024. These periods were selected based on the availability and type of data collected and the timing of remediation phases (table 2). Detailed methods for data collection and statistical analyses are provided below.

Data Collection Methods

Plymouth (U), Oak Grove (R), and Knox (D) are USGS streamgages with continuous water-quality monitors. Continuous gage height was recorded per USGS protocols and used to compute estimated discharge values for every 15-minute gage height value (Kennedy, 1984, Turnipseed and Sauer, 2010). The closest 15-minute discharge value for each of the three streamgage sites was used for associated discrete samples. At Upas Rd (U) and CR1100E (R), which are not co-located with USGS streamgages, discrete discharge measurements were made during each sampling event. The only exception occurred on June 18, 2024, when an equipment malfunction prevented a discharge measurement at CR1100E (R). For that date, discharge was estimated using data from the Oak Grove (R) streamgage because the difference in drainage area between the two sites is minimal. Continuous and discrete discharge data are available through the USGS National Water Information System —Water Data for the Nation (USGS, 2025).

Continuous water-quality was monitored at Plymouth (U), Oak Grove (R), and Knox (D), (table 1, fig. 3) following USGS protocols (Wagner and others, 2006). Water temperature and turbidity were measured using YSI EXO instruments (Xylem Incorporated, Yellow Springs, OH, YSI Inc., 2024), but only turbidity data were used in the analysis presented in this report. Turbidity is a measure of water clarity, often used as an indicator of in-stream suspended-sediment conditions. At Oak Grove (R) and Knox (D), water-quality monitoring of temperature and turbidity using a YSI EXO sensor started in 2019. At Plymouth (U) the water-quality monitor was deployed in February 2024, before expanded discrete sampling began at all five sites.

Discrete water-quality samples were collected 20 times at the five sites along the Yellow River between February 29 and October 23, 2024. Samples were collected using nationally

consistent field protocols for equal-width-interval sampling described in the USGS National Field Manual for the Collection of Water-Quality Data (USGS, 2006) and submitted to USGS laboratories for nutrient and sediment analyses (table 4). Samples were collected approximately biweekly, providing coverage for a wide range of discharges (fig. 4). On each sampling date, all five sites were sampled on the same day to capture relative hydrologic conditions. For each sample, the entire cross-section of the stream was sampled twice, either sequentially or concurrently depending on conditions, to produce one composite sample for nutrient analysis and one composite sample for sediment analysis. All discrete water-quality results were reviewed and quality assured by USGS personnel. The discrete water-quality results are publicly available through the Water Quality Portal (USGS and others, 2024).

Table 4. Constituents analyzed and the analytical methods for discrete water-quality samples collected during the Yellow River remediation study, northwest Indiana.

[LRL, laboratory reporting level; N, nitrogen; μm , micrometers; mg/L, milligram per liter; NWQL, U.S. Geological Survey National Water Quality Laboratory; <, less than; mL, milliliter; H_2SO_4 , sulfuric acid; P, phosphorus; EPA, U.S. Environmental Protection Agency; mm millimeter; n.a., not applicable]

Constituent	Field Preservation Method	Analysis Method	Method reference	LRL	Units	Laboratory
Sediment						
Suspended-sediment concentration	n.a	Filtration	Shreve and Downs, 2005	1	mg/L	Kentucky Sediment Laboratory
Nutrients						
Total Nitrogen as N, water, unfiltered	No filtration, acidify to pH <2 with 1 mL 4.5 normality H_2SO_4 , chill	Alkaline persulfate digestion	Patton and Kryskalla, 2003	0.1	mg/L	NWQL
Nitrate+nitrite as N, water, filtered	0.45- μm filter, chill	Colorimetry by enzymatic reduction	Patton and Kryskalla, 2011	0.08	mg/L	NWQL
Total ammonia as N, water, filtered	0.45- μm filter, chill	Colorimetry, Salicylate-Hypochlorite, Automated-Segment-ed Flow	Fishman, 1993	0.02	mg/L	NWQL
Nitrite as N, water, filtered	0.45- μm filter, chill	Colorimetry, Diazotization, Auto-Segment-ed Flow	Fishman, 1993	0.002	mg/L	NWQL
Phosphorus, as P, water, unfiltered	No filtration, acidify to pH <2 with 1 mL 4.5 normality H_2SO_4 , chill	Semiautomated colorimetry	US EPA, 1993 Method 365.1	0.008	mg/L	NWQL
Orthophosphate as P, water, filtered	0.45- μm filter, chill	Colorimetry, phosphomolybdate, automat-ed-segmented flow	Fishman, 1993	0.008	mg/L	NWQL

Quality-control samples for discrete water quality were collected throughout the study period. Two blank samples and two replicate samples were collected for nutrients, and four replicate samples were collected for suspended sediment at various sites and on different sampling dates. Quality-control samples were collected and processed concurrently with environmental samples to assess potential bias and variability associated with sample collection, handling, storage, shipping, and analysis (Mueller and others, 2015). All blank results were below applicable detection limits, and all replicate results were within acceptable relative percent differences, indicating that no bias or variability were introduced by the sampling and analysis processes.

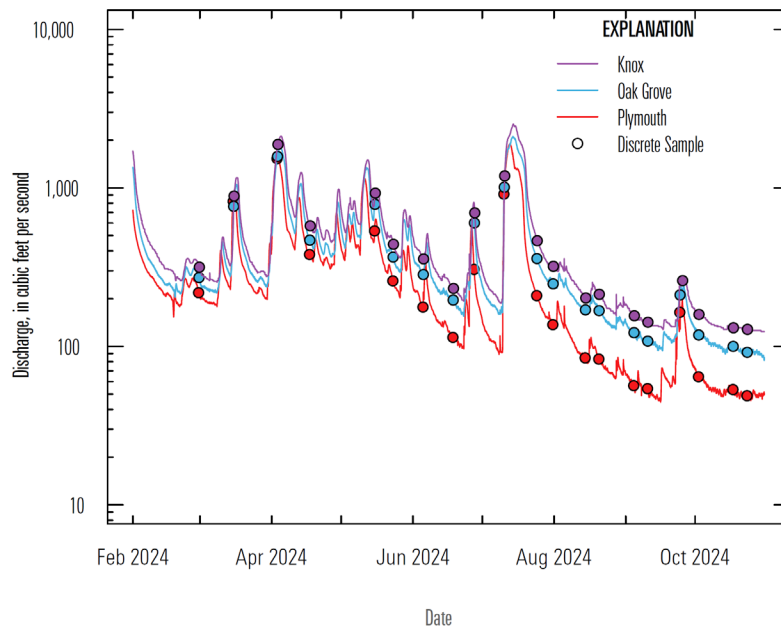


Figure 4. Discharge hydrographs for the post-remediation study period and discrete sampling times for the Yellow River sites at Plymouth, Oak Grove, and Knox, northwest Indiana.

Statistical Methods

Statistical methods used in this descriptive analysis included Generalized Least Squares (GLS) regression and Linear Mixed-Effects (LME) models. Discharge and water-quality data were transformed using the natural logarithm prior to analysis to satisfy assumptions of normality and homoscedasticity. GLS models are well-suited for datasets where the residuals are serially correlated, such as continuous time-series data. For this study, continuous time-series data included high-frequency turbidity and discharge measurements. To address the strong serial correlation inherent in continuous datasets, GLS models used the `nlme` package (Pinheiro and others, 2023). Rather than assigning a single correlation structure to all data, multiple candidate structures, including first- and second-order autoregressive (AR(1), AR(2)) and autoregressive moving average (ARMA), were evaluated. The final structure for each model was selected based on the lowest Akaike Information Criterion (AIC) and evaluation of residual diagnostics.

Conversely, LME models were used to analyze discrete water-quality samples collected repeatedly at the same sites over time. LME models used the `lme4` package (Bates and others, 2015), with degrees of freedom and *p*-values calculated using Satterthwaite's approximation in the `lmerTest` package (Kuznetsova and others, 2017). These models included both fixed effects representing factors of primary management interest and random effects accounting for dependence among repeated or grouped observations. Sampling date was included as a random effect to account for these repeated measurements, providing unbiased estimates of fixed effects on concentration and flux. Tables 5 and 6 specify the model parameters used in each analysis.

All statistical analyses were conducted in R (R Core Team, 2024). Post hoc comparison tests were performed using the `emmeans()` and `emtrends()` functions in the `emmeans`

package (Lenth, 2025). To aid interpretation, marginal means and slopes were calculated, and the resulting Estimated Marginal Means (EMMs) and pairwise comparisons were back-transformed from the log scale to the original observation units for reporting.

General Statistical Approach

Statistical tests were organized to address the two primary study questions (tables 5 and 6).

Question 1: Did water quality differ before and after remediation?

- 1a. GLS models were used to compare turbidity- discharge relations to determine differences in turbidity- discharge slope and turbidity values between periods at Oak Grove (R) and Knox (D).
- 1b. GLS models were used to compare discharge between periods at Oak Grove (R) and Knox (D).
- 1c. LME models were used to compare discrete suspended-sediment concentrations between periods at Oak Grove (R) and Knox (D).

Research Question 2: Did water quality differ among sites after remediation?

- 2a. GLS models were used to compare the turbidity- discharge relations to determine differences in turbidity- discharge slope and turbidity values between periods among sites at Plymouth (U), Oak Grove (R), and Knox (D).
- 2b. GLS models were used to compare discharge at Plymouth (U), Oak Grove (R), and Knox (D)
- 2c. LME models were used to compare mean nutrient and suspended-sediment concentrations and fluxes among Plymouth (U), Upas Rd (U), Oak Grove (R), CR1100E (R), and Knox (D).

- 2d. LME models were used to evaluate upstream-to-downstream monotonic trends in nutrient and suspended-sediment concentrations and fluxes at Plymouth (U), Upas Rd (U), Oak Grove (R), CR1100E (R), and Knox (D).

Comparison of Water Quality Before and After Remediation

GLS models were used to evaluate whether turbidity- discharge relations differed between the pre- and post-remediation periods at Oak Grove (R) and Knox (D) (table 5, test 1a). Models were defined with daily mean turbidity as the response variable, daily mean discharge as a continuous covariate, and remediation period (before and after) as a fixed effect. A discharge-by-period interaction term was included to test whether slopes differed between periods ($p < 0.05$). Pairwise t-tests of the resulting EMMs were used to assess differences in mean turbidity between periods while holding discharge constant at its mean value, and differences in marginal slopes between periods were evaluated using linear contrasts.

Table 5. Statistical tests used to address the research question, “Did water quality differ before and after remediation?” for the Yellow River, northwest Indiana. Site short names are defined in table 1.

[R, R Core Team, 2024; GLS, generalized least square; LME, linear mixed effects; ARMA, Autoregressive moving average; p, order of the autoregressive component; q, order of the moving-average component; ~, separates the response variable from explanatory variables; +, additive effects; *, main effects and their interaction; (1|month), random intercept for month; day_num, sequential daily time index; week_num, sequential weekly time index; —, none]

Test	Specific Research Question	Statistical Test Type	Response Variable ^a	Fixed Effects ^b	Random Effects	Correlation Structure	Sites	General Model Form in R
1a	Did the turbidity-discharge relations differ between periods?	GLS regression	Mean daily turbidity	Period, Discharge, Period × Discharge	—	ARMA(1,1)	Oak Grove, Knox ^c	Turbidity ~ Period*Discharge, correlation = corARMA(form = ~day_num, p = 1, q = 1)
1b	Did discharge differ between periods?	GLS regression	Mean weekly discharge	Period	—	ARMA(4,0)	Oak Grove, Knox	Discharge ~ Period, correlation = corARMA(form = ~week_num, p = 4, q = 0)

1c	Did suspended-sediment concentration differ between periods?	LME regression	Suspended-sediment concentration	Period, Discharge	Month	—	Oak Grove, Knox	Suspended Sediment Concentration ~ Period + Discharge + (1 month)
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^a Response variables were transformed using the natural logarithm.

^b Period and discharge were included as fixed effects where indicated; Period x Discharge represents the interaction between period and discharge

^c Oak Grove and Knox analyzed separately for test 1a.

To contextualize the turbidity results, GLS models were also used to test whether mean discharge differed between periods (table 5, test 1b). Models were defined with weekly mean discharge as the response variable and remediation period as a fixed effect. Weekly averages were used at both sites to ensure statistical robustness across the available period of record.

LME models were used to determine whether discrete suspended-sediment concentrations differed significantly between periods at Oak Grove (R) and Knox (D) (table 5, test 1c). These LME models were defined with suspended-sediment concentration as the response variable, discharge as a continuous covariate to flow-normalize the data, and remediation period as the primary fixed effect.

Comparison of Water Quality Among Sites After Remediation

GLS models were used to compare turbidity- discharge relations among Plymouth (U), Oak Grove (R), and Knox (D) during the post-remediation period (table 6, test 2a). Models defined daily mean turbidity as the response variable, daily mean discharge as a continuous covariate, and site as a fixed effect. A discharge-by-site interaction term was included to test whether turbidity-discharge slopes differed among sites ($p < 0.05$). Post hoc comparisons were used to evaluate differences in estimated mean turbidity and marginal slopes among sites. GLS models were also used to evaluate whether daily mean discharge differed among Plymouth (U),

Oak Grove (R), and Knox (D) during the post-remediation period, with site included as the fixed effect (table 6, test 2b).

LME models were used to assess whether discrete concentrations and fluxes of suspended sediment, total nitrogen, nitrate plus nitrite, nitrite, total ammonia, total phosphorus, and orthophosphate differed among Plymouth (U), Upas Rd (U), Oak Grove (R), CR1100E (R), and Knox (D) (table 6, test 2c). For models evaluating concentration, discharge was included as a covariate to flow-normalize the data. For models evaluating flux, season (winter, spring, summer, and fall) replaced discharge to account for seasonal hydrologic variability. Site was included as the primary fixed effect in all concentration and flux models.

Table 6. Statistical tests used to address the research question, "Did water quality differ among sites after remediation?" for the Yellow River, northwest Indiana. Site short names are defined in table 1.

[R, R Core Team, 2024; GLS, generalized least square; LME, linear mixed effects; —, none; ARMA, autoregressive moving average; AR(1), first order autoregressive; p, order of the autoregressive component; q, order of the moving-average component; ~, separates the response variable from explanatory variables; +, additive effects; *, main effects and their interaction; (1|date), random intercept of sampling date; datenum, *sequential daily time index*; weeknum, *sequential weekly time index*; site_num, ordered numeric variable representing sites from upstream to downstream; —, none]

Test	Specific Research Question	Statistical Test Type	Sites	Response Variable ^a	Fixed Effects	Random Effects	Correlation Structure	General Model Form in R
2a	Did the turbidity-discharge relations differ among Plymouth, Oak Grove, and Knox post-remediation?	GLS model	Plymouth, Oak Grove, Knox	Daily mean turbidity	Site, Discharge, Site × Discharge	—	ARMA(1,1)	Turbidity ~ Site*Discharge, correlation = corARMA(form = ~ date_num, p = 1, q = 1)
2b	Did discharge differ among Plymouth, Oak Grove, and Knox in post-remediation?	GLS model	Plymouth, Oak Grove, Knox	Weekly mean discharge	Site	—	AR(1)	Discharge ~ Site, correlation = corAR1(form = ~ week_num)
2c	Did the means of water-quality parameters differ among sites?	LME model	Plymouth, Upas Rd, Oak Grove, CR1100E, Knox	Multiple parameters ^b	Site (factor)	Sampling Date	—	Response ~ Site + (1 date)
2d	Was there a monotonic upstream-to-downstream trend in water quality?	LME model	Plymouth, Upas Rd, Oak Grove, CR1100E, Knox	Multiple parameters ^b	Site position (ordered numeric) ^c	Sampling Date	—	Response ~ Site_num + (1 date)

^a Response variables were transformed using the natural logarithm.

^b Parameters include concentrations and fluxes of suspended sediment, total nitrogen (as N), nitrite+nitrate (as N), total ammonia (as N), nitrite (as N), phosphorus (as P), orthophosphate (as P), and discharge.

^c Site position was represented as an ordered numeric variable increasing from upstream to downstream to test for a monotonic longitudinal trend

Log-likelihood ratio tests (LRTs) were used to compare full models (including site) to null models (excluding site) to evaluate the overall significance of the site effect ($p < 0.05$). These tests were conducted using maximum likelihood (ML) estimation to allow for direct comparison of models with differing fixed effects. All models were then fit using Restricted Maximum Likelihood (REML) to obtain unbiased variance estimates and generate the final EMMs used for pairwise comparisons among sites.

Separate LME models were used to evaluate whether concentrations and fluxes exhibited a monotonic upstream-to-downstream trend during the post-remediation period (table 6, test 2d). Sites were assigned ordered numeric values from 1 for Plymouth (U), the most upstream site, to

5 for Knox (D), the most downstream site. Concentration models were flow-normalized using discharge as a covariate, while flux models included season to control for seasonal hydrologic variability. Site order was included as the fixed effect in all models. LRTs were used to compare full and null models to determine whether the linear effect of site order was significant ($p < 0.05$). A significant model slope indicated whether the upstream-to-downstream monotonic trend was increasing (positive slope) or decreasing (negative slope).

Results and Discussion

The effects of remediation efforts in the Yellow River were evaluated by addressing two overarching questions: (1) Did water quality differ before and after remediation? and (2) Did water quality differ among sites after remediation? Results of the statistical tests are presented in tables 7 –11, and additional statistical output summaries are provided in Appendix 1.1 –1.5.

Did water quality differ before and after remediation?

Comparison of turbidity- discharge relations before and after remediation (test 1a) indicated significantly lower turbidity at a given discharge during the post-remediation period at both Oak Grove (R) and Knox (D) (fig. 5; table 7). In contrast, comparisons of overall mean daily discharge (test 1b), mean daily turbidity (test 1a), and suspended-sediment concentrations (test 1c) showed that none of these metrics were different between periods at either site (fig. 6; table 7).

These findings indicate that turbidity was less responsive to changes in discharge after remediation at both Oak Grove (R) and Knox (D) (test 1a), suggesting that remediation efforts may have influenced turbidity dynamics at sites located within and downstream from the remediated reach. A previous study of restored agricultural ditches in Indiana and Ohio also

found lower turbidity in restored reaches compared to upstream, channelized controls (Mahl and others, 2015). Similarly, work in experimental watersheds in New Zealand showed that riparian restoration reduced in-stream turbidity (Collins and others, 2013). A USGS study in New York also documented reductions in suspended-sediment concentrations following implementation of sediment and turbidity reduction projects (STRPs), such as floodplain reconnection and hillslope stabilization (Siemion and others, 2016). Although the Yellow River data indicate initial changes in turbidity-discharge relations after remediation, additional post-remediation data could help determine whether these changes persist over time and whether longer-term changes in mean turbidity and suspended-sediment concentrations become evident.

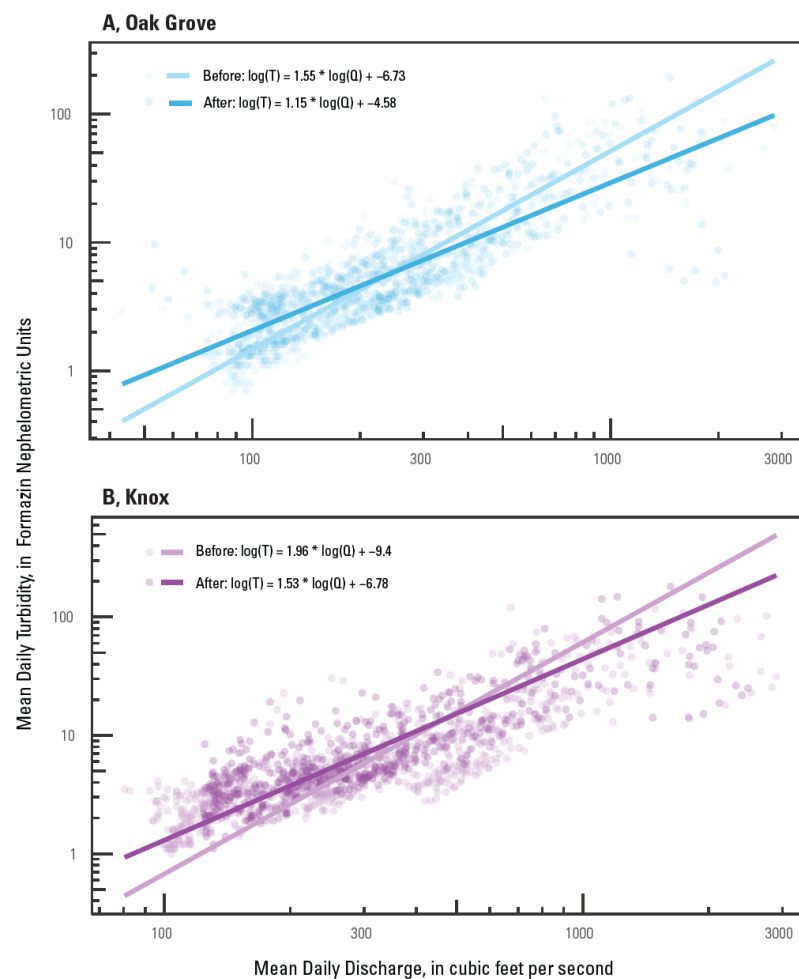


Figure 5. Discrete sample data plotted for the turbidity (T) to discharge (Q) relationship at (a) Oak Grove and (b) Knox, with generalized least squares regression model lines for the pre-remediation and post-remediation periods.

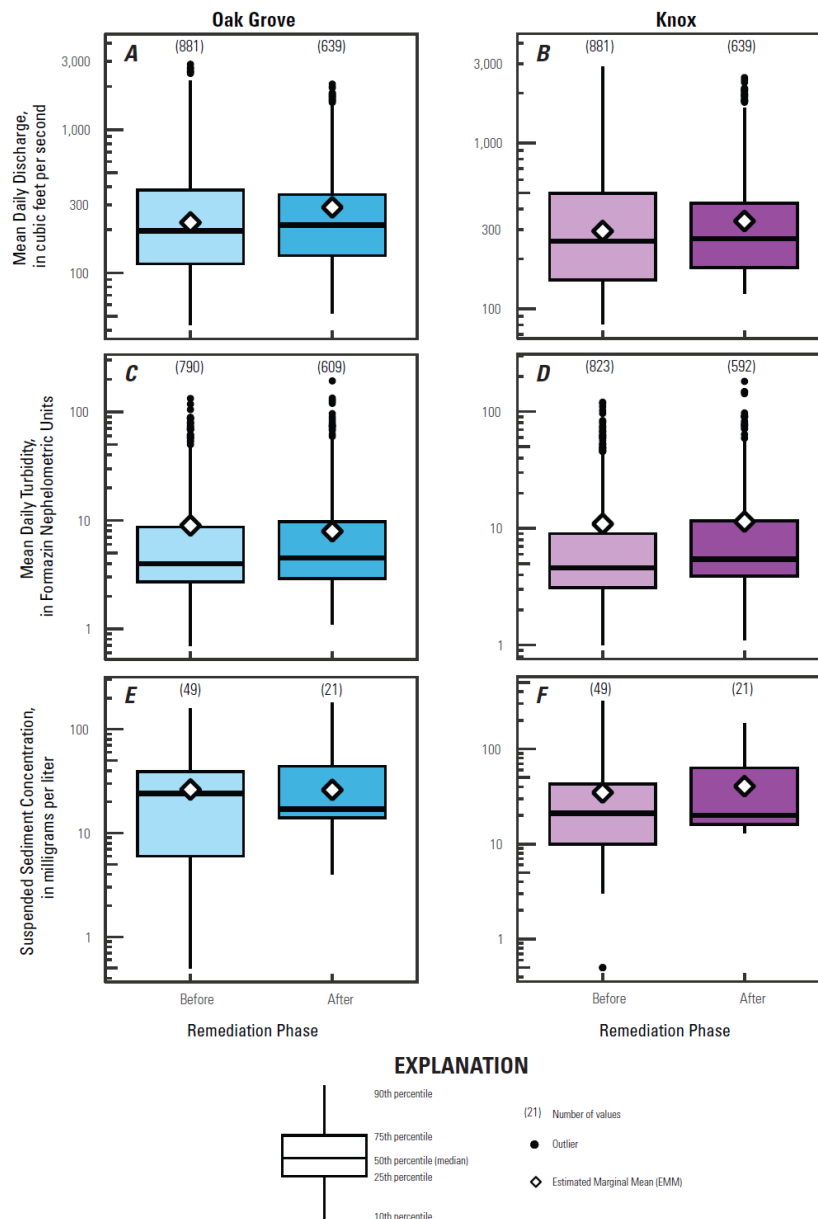


Figure 6. Boxplots showing the data for (A-B), daily mean discharge; (C-D), daily mean turbidity; and (E-F), discrete suspended-sediment concentrations before and after remediation at Oak Grove and Knox. Boxplots show the distribution and median of the observed data. Diamonds represent estimated marginal means for each variable as a result of statistical tests; generalized least square regression for discharge and turbidity and linear mixed-effects regression for suspended-sediment concentrations.

Table 7. Results of comparisons of turbidity-discharge slopes and estimated marginal means for discharge, turbidity, and suspended-sediment concentrations at two Yellow River sites in northwest Indiana before and after remediation.

[<, less than; FNU, Formazin Nephelometric Units, Data used in analysis available at U.S. Geological Survey, 2025. Tests described in table 5]

Test	Response Variable	Site							
		Oak Grove				Knox			
		Before Remediation	After Remediation	p - value	Significant Difference?	Before Remediation	After Remediation	p - value	Significant Difference?
1a	Turbidity-discharge slope	1.55	1.15	<0.001	Yes	1.96	1.53	<0.001	Yes
1a	Turbidity, in FNU	8.96	7.92	0.30	No	10.90	11.40	0.78	No
1b	Discharge, in cubic feet per second	225	287	0.52	No	294	338	0.70	No
1c	Suspended-sediment concentration, in milligrams per liter	26.2	26.0	0.97	No	34.70	40.50	0.42	No

Did water quality differ among sites after remediation?

Comparison of turbidity- discharge relations among sites during the post-remediation period (test 2a) indicated that slopes at Plymouth (U) and Oak Grove (R) were similar to Knox (D), whereas the slope at Oak Grove (R) was significantly steeper than at Plymouth(U) (fig. 7C; table 8). Comparison of discharge among sites post-remediation (test 2b) indicated that estimated mean weekly discharge did not differ significantly among the three sites (fig. 7A; table 8). In contrast, estimated mean daily turbidity was significantly lower at Oak Grove (R) and Knox (D) than at Plymouth (U) (fig. 7B, table 8).

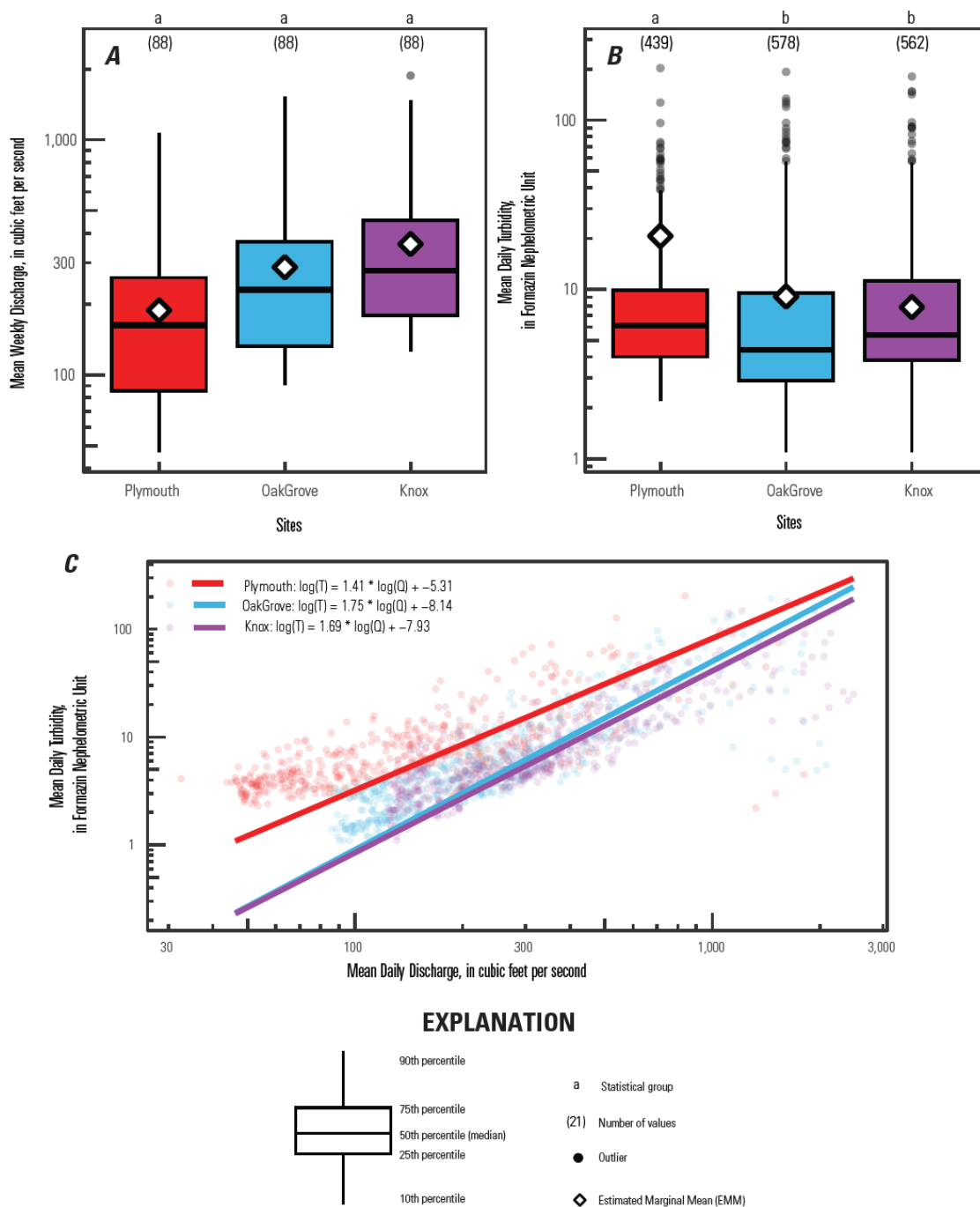


Figure 7. Differences in discharge, turbidity, and turbidity- discharge relations at Plymouth, Oak Grove, and Knox during the post-remediation period of the Yellow River remediation study, northwest Indiana. (A) mean weekly discharge; (B) mean daily turbidity; (C) relation between mean daily turbidity and mean daily discharge. For A and B, discrete sample data are used for depicted boxplots,

and diamonds represent estimated marginal means from generalized least-squares models. Lower case letters, a and b, above boxplots indicate statistically significant differences ($p < 0.05$) between groups. For C, discrete sample data is overlain by generalized least squares regression lines for each site.

Table 8. Comparison of turbidity- discharge slopes, mean daily turbidity, and mean daily discharge among three sites (Plymouth, Oak Grove, and Knox) on the Yellow River during the post-remediation period. Estimates and standard errors, as well as significant differences among sites found using post hoc pairwise comparison tests are presented. Sites with statistically significant differences ($p < 0.05$) are indicated by different lowercase letters.

[+/-, plus or minus; a, b, significantly different group; FNU, Formazin Nephelometric Units. Site short names are defined in table 1. Data used in analysis available at U.S. Geological Survey, 2025. Tests described in table 6]

Test	Response Variable	Estimates by site		
		Plymouth	Oak Grove	Knox
Test 2a	Turbidity-discharge relation, Slope estimate	1.41 <i>a</i> (+/- 0.09)	1.75 <i>b</i> (+/- 0.09)	1.69 <i>ab</i> (+/- 0.10)
Test 2a	Mean daily turbidity, in FNU	20.68 <i>a</i> (+/- 4.74)	9.08 <i>b</i> (+/- 1.90)	7.86 <i>b</i> (+/- 1.63)
Test 2b	Mean weekly discharge, in cubic feet per second	188 <i>a</i> (+/- 78)	287 <i>a</i> (+/- 119)	360 <i>a</i> (+/- 150)

Post-remediation comparisons of turbidity- discharge relations indicated that turbidity at Oak Grove (R) and Knox (D) was more responsive to changes in discharge than at Plymouth (U) (test 2a). At first, this result may initially appear inconsistent with turbidity-discharge comparisons from the before-after analysis described above (test 1a); however, it is important to note that before-after results capture long-term patterns in water-quality response following remediation at Oak Grove (R) and Knox (D), while post-remediation results provide only a snapshot comparison of site-level water-quality differences among sites upstream, within, and downstream of remediation. The finding that mean turbidity was lower at Oak Grove (R) and Knox (D) compared to Plymouth (U) post remediation (test 2a) is consistent with improved water clarity within and downstream of the remediation reach.

Turbidity is frequently used to assess spatial and temporal patterns in water quality. For example, Schlosser and Karr (1981) found higher turbidity in stream sites near highly erosive

areas, and Jabbar and Grote (2019) reported seasonal differences in turbidity due to spring runoff in an agricultural watershed. Others have demonstrated that turbidity responds to both in-stream and watershed-scale changes. Shields and others (2009) noted the influence of channel incision on turbidity in an urban stream while Davis and others (2015) and Mahl and others (2015) found that floodplain restoration via two-stage ditch construction reduced turbidity in agricultural streams relative to channelized reaches. At the landscape scale, researchers have found that turbidity increases with agricultural land use in watersheds (Galbraith and Burns, 2006; Gunawardana and MacDonald, 2024), though management changes like planting cover crops at the watershed-scale have also been shown to reduce turbidity, particularly during high-precipitation events (Trentman and others, 2022). These studies demonstrate the utility of turbidity as an integrative measure of spatial and temporal changes in stream water quality and lend support to the interpretation that results here indicate initial improvements in water-quality in the Yellow River following streambank remediation.

Comparison of mean concentrations among sites during the post-remediation period indicated differences in several water-quality constituents (test 2c; fig. 8, table 9). Discharge was greatest at Knox (D), intermediate at CR1100E (R), Oak Grove (R), and Upas Rd (U), and lowest at Plymouth (U) (fig. 8A). Mean suspended-sediment concentration was numerically lower at all sites downstream from Plymouth (U), but the difference was statistically significant only at Oak Grove (R) (fig. 8B). Mean concentrations of total nitrogen (fig. 8C), nitrate plus nitrite (fig. 8D), ammonia (fig. 8F), and total phosphorus (fig. 8G) were lowest at Knox (D), intermediate at Upas (U), Oak Grove (R), and CR1100E (R), and greatest at Plymouth (U) (fig. 8A). Mean nitrite concentrations were significantly lower at all sites than at Plymouth (U) (fig.

8E), whereas mean orthophosphate concentrations were lowest at Knox (D) and similar among the other sites (fig 8H).

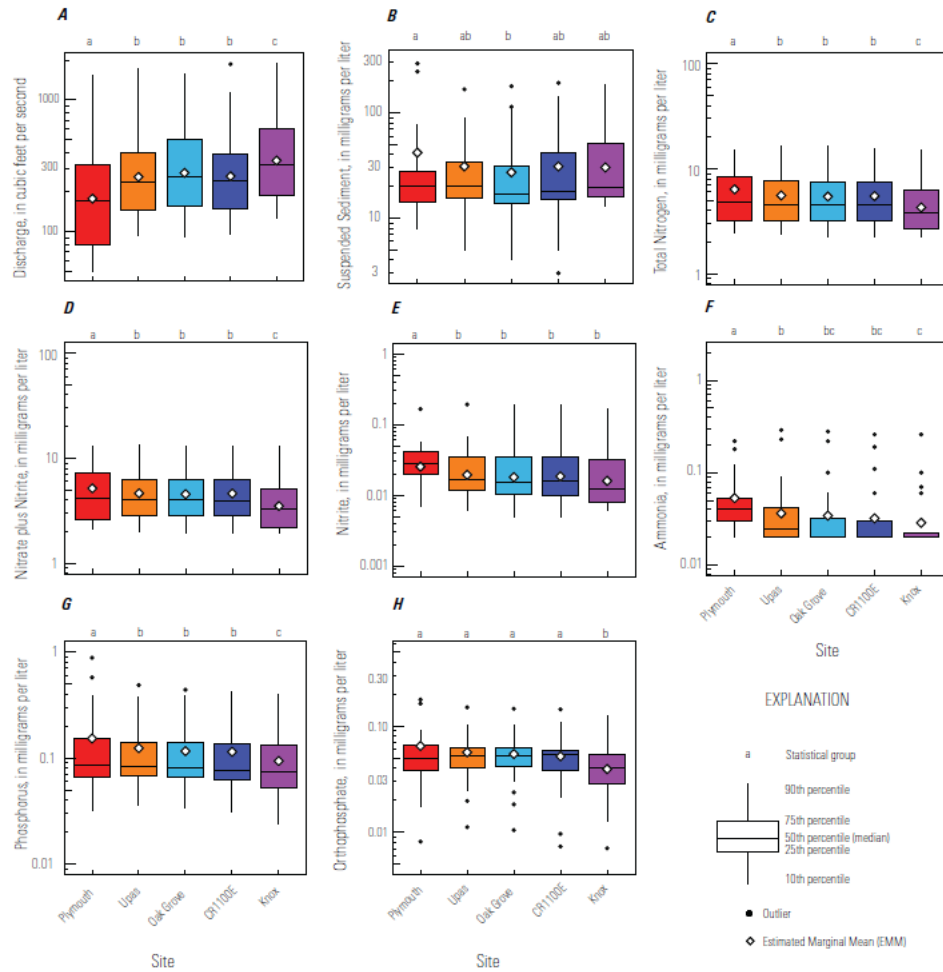


Figure 8. Boxplots of discrete measurements of A, discharge; B, suspended sediment; C, total nitrogen; D, nitrate plus nitrite; E, nitrite; F, ammonia; G, phosphorus; H, orthophosphate along Yellow River at Plymouth, Upas Rd, Oak Grove, CR1100E, and Knox during the post-remediation period. Diamonds represent estimated marginal means from linear mixed-effects models. Groups with statistically significant differences ($p < 0.05$) are indicated by different lowercase letters.

Table 9. Estimated marginal mean discharge and concentrations of suspended sediment and nutrients measured in discrete water-quality samples collected at five sites along the Yellow River, northwest Indiana during the post-remediation period. Linear mixed-effects models were used to evaluate differences in mean concentrations among sites. Groups with statistically significant differences ($p < 0.05$) are indicated by different lowercase letters.

[N, nitrogen; P, phosphorus; CR1100E, Rd, Road; County Road 1100 East; $\pm$, plus or minus; *a, b, c*, significantly different groups. Data used in analysis available at U.S. Geological Survey, 2025.]

Constituent	Site (Upstream to Downstream)				
	Plymouth	Upas Rd	Oak Grove	CR1100E	Knox
Discharge					
Discharge, in cubic feet per second	178 $\pm$ 32.8 <i>a</i>	260 $\pm$ 47.9 <i>b</i>	279 $\pm$ 51.5 <i>b</i>	263 $\pm$ 48.7 <i>b</i>	346 $\pm$ 63.8 <i>c</i>
Sediment					
Suspended-sediment concentration, in milligrams per liter	41.8 $\pm$ 7.50 <i>a</i>	31.0 $\pm$ 4.83 <i>ab</i>	27.2 $\pm$ 4.15 <i>b</i>	31.0 $\pm$ 4.85 <i>ab</i>	30.3 $\pm$ 4.43 <i>ab</i>
Nitrogen					
Total nitrogen as N, in milligrams per liter	6.38 $\pm$ 0.57 <i>a</i>	5.59 $\pm$ 0.47 <i>b</i>	5.44 $\pm$ 0.46 <i>b</i>	5.51 $\pm$ 0.46 <i>b</i>	4.29 $\pm$ 0.35 <i>c</i>
Nitrate + nitrite as N, in milligrams per liter	5.17 $\pm$ 0.47 <i>a</i>	4.64 $\pm$ 0.40 <i>b</i>	4.54 $\pm$ 0.39 <i>b</i>	4.61 $\pm$ 0.40 <i>b</i>	3.51 $\pm$ 0.30 <i>c</i>
Nitrite as N, in milligrams per liter	0.03 $\pm$ 0.01 <i>a</i>	0.02 $\pm$ 0.00 <i>b</i>	0.02 $\pm$ 0.00 <i>b</i>	0.02 $\pm$ 0.00 <i>b</i>	0.02 $\pm$ 0.00 <i>b</i>
Total ammonia as N, in milligrams per liter	0.05 $\pm$ 0.01 <i>a</i>	0.04 $\pm$ 0.01 <i>b</i>	0.03 $\pm$ 0.01 <i>bc</i>	0.03 $\pm$ 0.01 <i>bc</i>	0.03 $\pm$ 0.00 <i>c</i>
Phosphorus					
Total phosphorus as P, in milligrams per liter	0.15 $\pm$ 0.02 <i>a</i>	0.12 $\pm$ 0.01 <i>b</i>	0.12 $\pm$ 0.01 <i>b</i>	0.11 $\pm$ 0.01 <i>b</i>	0.09 $\pm$ 0.01 <i>c</i>
Orthophosphate as P, in milligrams per liter	0.06 $\pm$ 0.01 <i>a</i>	0.06 $\pm$ 0.01 <i>a</i>	0.05 $\pm$ 0.01 <i>a</i>	0.05 $\pm$ 0.01 <i>a</i>	0.04 $\pm$ 0.01 <i>b</i>

Flux incorporates both constituent concentration and stream discharge to quantify the mass of a constituent transported past a location over a specified period. Because flux accounts for the volume of water moving through the system, it provides a meaningful basis for comparing upstream and downstream conditions and evaluating how loads change across a watershed. Comparisons of mean constituent fluxes among sites (test 2c) also indicated significant differences (fig. 9; table 10), although flux patterns differed from concentration patterns. For example, mean suspended-sediment flux was significantly greater at Knox (D) than all other sites (fig. 9A). Mean flux of total nitrogen (fig. 9B), nitrate plus nitrite (fig. 9C), nitrite

(fig. 9D), total phosphorus (fig. 9F), and orthophosphate (fig. 9G) were similar at Knox (D), CR1100E (R), Oak Grove (R), and Upas Rd (U) and were significantly greater at those sites than at Plymouth (U). Mean ammonia flux did not differ significantly among sites (fig. 9E).

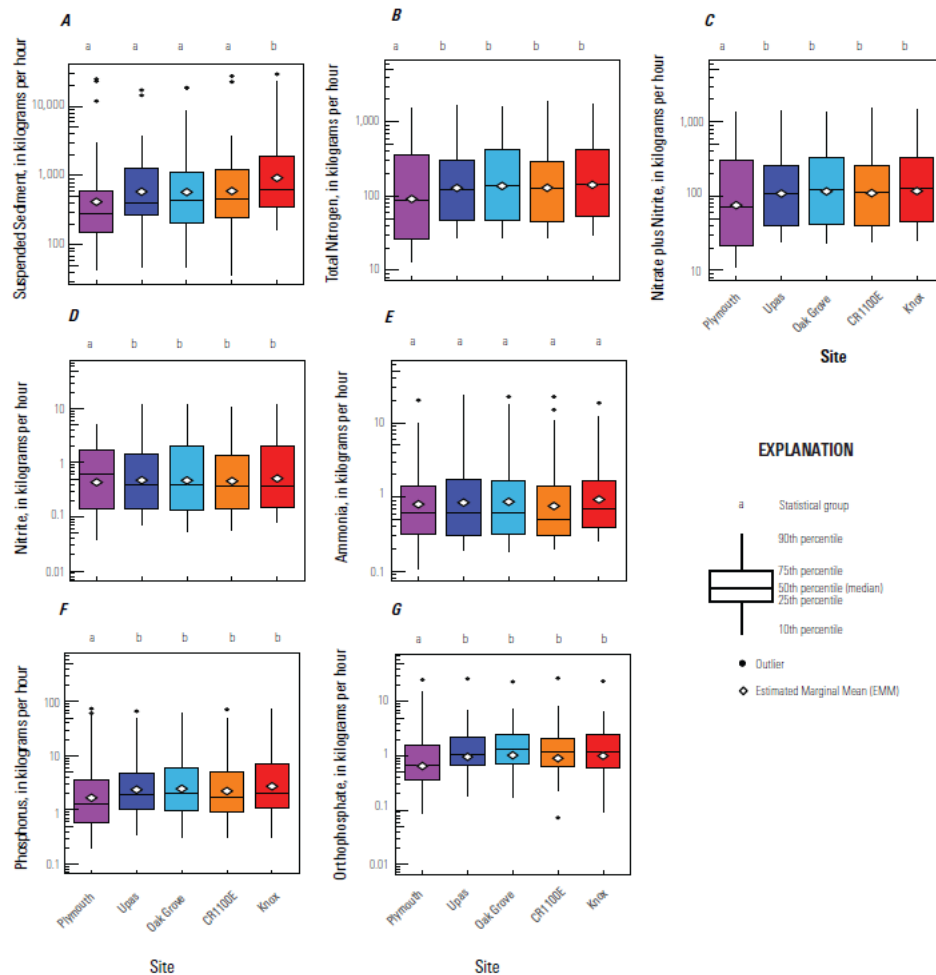


Figure 9. Boxplots of instantaneous flux calculated from discrete sample results for A, suspended sediment; B, total nitrogen; C, nitrate plus nitrite; D, nitrite; E, ammonia; F, phosphorus; and G, orthophosphate along Yellow River at Plymouth, Upas Rd, Oak Grove, CR1100E, and Knox during the post-remediation period. Diamonds represent estimated marginal means from linear mixed-effects models. Groups with statistically significant differences ($p < 0.05$) are indicated by different lowercase letters.

Table 10. Estimated marginal mean instantaneous fluxes of suspended sediment and nutrients measured in discrete water-quality samples collected at five sites along the Yellow River, northwest Indiana during the post-remediation period. Linear mixed-effects models were used to evaluate differences in mean loads among sites. Sites with statistically significant differences ($p < 0.05$) are indicated by different lowercase letters.

[N, nitrogen; P, phosphorus; Rd, Road; CR1100E, County Road 1100 East; $\pm$, plus or minus; *a*, *b*, *c*, significantly different groups. Site short names from table 1. Data used in analysis available at U.S. Geological Survey 2025.]

Instantaneous constituent flux, in kilograms per hour	Site (Upstream to Downstream)				
	Plymouth	Upas Rd	Oak Grove	CR1100E	Knox
Sediment					
Suspended sediment	419 $\pm$ 159 <i>a</i>	586 $\pm$ 222 <i>a</i>	578 $\pm$ 219 <i>a</i>	599 $\pm$ 228 <i>a</i>	925 $\pm$ 351 <i>b</i>
Nitrogen					
Total nitrogen as N	91.2 $\pm$ 22.4 <i>a</i>	128 $\pm$ 31.6 <i>b</i>	137 $\pm$ 33.6 <i>b</i>	129 $\pm$ 31.8 <i>b</i>	141 $\pm$ 34.8 <i>b</i>
Nitrate + nitrite as N	75.2 $\pm$ 17.6 <i>a</i>	108 $\pm$ 25.3 <i>b</i>	115 $\pm$ 27.0 <i>b</i>	109 $\pm$ 25.7 <i>b</i>	117 $\pm$ 27.3 <i>b</i>
Nitrite as N	0.44 $\pm$ 0.14 <i>a</i>	0.48 $\pm$ 0.15 <i>a</i>	0.47 $\pm$ 0.15 <i>a</i>	0.46 $\pm$ 0.15 <i>a</i>	0.51 $\pm$ 0.16 <i>a</i>
Total ammonia as N	0.80 $\pm$ 0.27 <i>a</i>	0.84 $\pm$ 0.29 <i>a</i>	0.86 $\pm$ 0.29 <i>a</i>	0.76 $\pm$ 0.26 <i>a</i>	0.93 $\pm$ 0.32 <i>a</i>
Phosphorus					
Total phosphorus as P	1.67 $\pm$ 0.59 <i>a</i>	2.36 $\pm$ 0.83 <i>b</i>	2.45 $\pm$ 0.86 <i>b</i>	2.22 $\pm$ 0.78 <i>b</i>	2.72 $\pm$ 0.96 <i>b</i>
Orthophosphate as P	0.65 $\pm$ 0.18 <i>a</i>	0.96 $\pm$ 0.27 <i>b</i>	1.03 $\pm$ 0.29 <i>b</i>	0.90 $\pm$ 0.26 <i>b</i>	1.00 $\pm$ 0.28 <i>b</i>

These results indicate that water quality varied among sites during the post-remediation period. Concentration patterns reflected site-specific differences, whereas flux patterns highlighted the overwhelming influence of greater discharge at downstream sites. Concentrations of most constituents were lower at sites within and downstream from the remediated reach — Oak Grove (R), CR1100E (R), and Knox (D) — than at Plymouth (U) but were generally similar to concentrations at Upas (U). This suggests that spatial proximity may have played a larger role than remediation status. Plymouth (U) is approximately 18 miles upstream from the first remediation area, whereas Upas Rd (U) is approximately 1 mile upstream of the remediation and is spatially closer to Oak Grove (R) and CR1100E (R). These three spatially clustered sites also had generally similar mean concentrations. Similarities among nearby sites may reflect shared watershed characteristics such as land use, soils, and topography. It is also possible that one year

of data collection was insufficient to detect the influence of remediation on mean concentrations, even when accounting for variation in discharge. Previous synoptic sampling studies in restored systems have reported similar longitudinal variability in water chemistry, noting the influence of local land use factors and groundwater inputs (Sivirichi and others, 2011). Because flux integrates both concentration and discharge, greater downstream fluxes are likely driven by higher discharge rather than differences in water quality.

Nevertheless, previous studies have shown that restoration can improve water quality, but effects can be temporally variable, site-specific, or more detectable in functional rather than concentration-based metrics. For example, two-stage floodplain restorations in agricultural streams reduced phosphorus concentrations during high-flow events (i.e., floodplain inundation), but the effects on sediment, nitrate, and ammonium concentrations were more variable (Davis and others, 2015; Mahl and others 2015). Restored reaches also have been shown to export less total nitrogen and phosphorus during high flows than unrestored reaches (Pennino and others, 2016), and long-term monitoring in the Chesapeake Bay watershed documented decreases in nitrate concentrations over a decade following restoration (Mayer and others, 2022). Numerous studies further demonstrate that restoration improves functional metrics such as nutrient uptake or retention, ultimately reducing downstream export (Sivirichi and others, 2011; Filoso and Palmer, 2011; Newcomer Johnson and others, 2016). Future research in the Yellow River could incorporate targeted storm sampling and functional metrics to more fully evaluate the effects of remediation on water quality.

Finally, examining monotonic trends in mean concentration and flux (test 2d) revealed upstream-to-downstream patterns for some water-quality parameters during the post-remediation period (table 11). Mean concentration of total nitrogen, nitrate, nitrite, ammonia, total

phosphorus, and orthophosphate decreased from upstream to downstream (table 11), whereas mean suspended-sediment concentration showed no monotonic trend. In contrast, mean fluxes of suspended sediment, total nitrogen, nitrate, total phosphorus, and orthophosphate increased from upstream to downstream (table 11), whereas mean nitrite and ammonia fluxes showed no significant trend.

These findings show contrasting longitudinal patterns between concentration and flux metrics. Nutrient concentrations decreased from upstream to downstream, whereas suspended sediment did not. Concentration models accounted for discharge, thus these longitudinal declines in nutrients, but not sediment, likely reflect mechanisms beyond simple dilution as water moves downstream, such as in-stream processing or localized remediation effects. In contrast, nutrient and sediment fluxes increased longitudinally, a pattern best explained by hydrologic scaling which is when larger volumes of flow downstream lead to greater mass export despite lower concentrations. Similar dynamics have been documented in restored systems, where nitrogen concentrations decreased downstream while flux increased (Filoso and Palmer, 2011). Collectively, the Yellow River results suggest that remediation may be effective at reducing concentrations locally, but increasing discharge downstream continues to drive nutrient and sediment fluxes. This highlights how evaluating restoration performance using concentration- and flux-based metrics and longitudinal trends within the stream can be used to evaluate potential benefits despite discontinuous remediation.

Table 11. Fixed-effect estimates from linear mixed-effect models evaluating the direction and significance of upstream-to-downstream trends in post-remediation sediment and nutrient concentrations and fluxes in the Yellow River, northwest Indiana.

[N, nitrogen; P, phosphorus; <, less than; Data used in analysis available at U.S. Geological Survey, 2025.]

Constituent	Estimate	Standard Error	P-value	Significant?	Trend Direction
Discharge, in cubic feet per second					
Discharge	0.13	0.02	< 0.01	Yes	Increasing downstream
Concentration, in milligrams per liter					
Suspended sediment	-0.05	0.03	0.09	No	No trend
Total nitrogen as N	-0.07	0.01	< 0.01	Yes	Decreasing downstream
Nitrate + nitrite as N	-0.07	0.01	< 0.01	Yes	Decreasing downstream
Nitrite as N	-0.07	0.02	< 0.01	Yes	Decreasing downstream
Total ammonia as N	-0.11	0.02	< 0.01	Yes	Decreasing downstream
Total phosphorus as P	-0.09	0.02	< 0.01	Yes	Decreasing downstream
Orthophosphate as P	-0.10	0.02	< 0.01	Yes	Decreasing downstream
Flux, in kilograms per hour					
Suspended sediment	0.16	0.03	< 0.01	Yes	Increasing downstream
Total nitrogen as N	0.09	0.02	< 0.01	Yes	Increasing downstream
Nitrate + nitrite as N	0.09	0.02	< 0.01	Yes	Increasing downstream
Nitrite as N	0.03	0.02	0.09	No	No trend
Total ammonia as N	0.02	0.02	0.34	No	No trend
Total phosphorus as P	0.09	0.02	< 0.01	Yes	Increasing downstream
Orthophosphate as P	0.08	0.03	< 0.01	Yes	Increasing downstream

Limitations and Considerations

Data collection along the Yellow River was expanded in February 2024 to support longitudinal analysis of remediation effects. However, comparisons between pre- and post-remediation conditions were limited to one site within the remediated reach and one site downstream from remediation. Pre-remediation data were not available at the upstream control site for turbidity, discharge, or suspended-sediment concentration, preventing comparison of

temporal changes among sites upstream from, within, and downstream from the remediated reach. Nutrient data also were limited to discrete samples collected in 2024, with no pre-remediation baseline, available for evaluating changes over time.

Sites where data was collected only post-remediation were chosen to encompass the immediate effect of remediation in terms of proximity to the remediation reach. Upas Rd (U), the site just upstream of remediation, and sites within the remediation reach, Oak Grove (R) and CR1100E (R), showed similar concentration and flux results. Remediation projects in the study area were completed between 2021 and 2023, and their full effects may not yet be realized. Other restoration studies (Siemion and others, 2016; Collins and others, 2013) have found that bank remediation effects become more pronounced as the restored riparian zones mature over time. Continued data collection could improve the ability to evaluate sediment and nutrient transport and determine whether longer-term changes become evident.

Summary

The U.S. Geological Survey, in cooperation with the Kankakee River Basin and Yellow River Basin Development Commission, evaluated the potential effects of streambank remediation along portions of the Yellow River in northwest Indiana. Two phases of remediation were completed between 2021 and 2023, and potential effects were evaluated through temporal and spatial comparisons. For the temporal analysis, two periods were defined: a pre-remediation period (April 2019 – August 2021) and a post-remediation period (February – October 2024). Continuous discharge, continuous turbidity, and discrete suspended-sediment data were available at two long-term monitoring sites—USGS 05516665 Yellow River near Oak Grove, IN (Oak Grove) and USGS 05517000 Yellow River at Knox, IN (Knox). These data were used to

evaluate changes in streamflow, turbidity, and suspended-sediment dynamics before and after remediation.

For the spatial analysis, continuous water-quality monitoring was expanded in February 2024 to include USGS 05516500 Yellow River at Plymouth, IN (Plymouth), where turbidity monitoring was added to the existing streamflow gage. In addition to the three continuous-monitoring sites at Plymouth, Oak Grove, and Knox, two supplemental sites located upstream of and within the remediation reach—USGS 05516663 Yellow River at Upas Road near Oak Grove, IN (Upas Rd) and USGS 05516667 Yellow River at CR1100E near Oak Grove, IN (CR1100E)—were sampled during the post-remediation period for suspended sediment and dissolved and total forms of nitrogen and phosphorus.

Statistical analysis in this study consisted of generalized least-squares and/or linear mixed-effects regression models, depending on the structure of the data. To address whether water quality differed after remediation (temporally), changes were analyzed in turbidity-discharge relations, and turbidity, discharge, and suspended-sediment concentrations between the pre- and post- remediation periods. Spatial differences during the post-remediation period were evaluated from upstream to downstream using turbidity, discharge, suspended-sediment concentrations and fluxes, and nutrient concentrations and fluxes.

Temporal comparisons at Oak Grove (R - within remediation), and Knox (D - downstream of remediation) showed that turbidity increased less with increasing discharge, indicating that water clarity during larger flow events improved after remediation compared to before remediation. Mean turbidity and mean suspended-sediment concentrations, however, did not show significant differences between remediation periods. These results suggest initial

improvements in turbidity-discharge relation, although additional long-term monitoring could help determine whether these changes persist or strengthen over time.

Upstream to downstream comparisons after remediation showed that turbidity -discharge relations had steeper slopes at downstream sites relative to the upstream control site, Plymouth (U), meaning turbidity increased more rapidly with discharge downstream. Oak Grove (R) and Knox (D) exhibited similar slopes, indicating comparable turbidity responses to changing flow conditions at these downstream locations. Despite stronger turbidity response to discharge, downstream sites within and downstream of remediation had statistically lower mean turbidity, further suggesting improved water clarity related to remediation. Nutrient concentrations generally decreased from upstream to downstream, whereas suspended-sediment concentration showed some decrease but with no consistent upstream to downstream trend. In contrast, fluxes of suspended sediment and nutrient constituents increased downstream, likely driven by greater discharge volumes.

This study was limited by the availability of pre-remediation data only at sites within and downstream from the remediated reach. Without pre-remediation data from an upstream control site, the observed temporal and spatial patterns could not be fully attributed to remediation. The lack of pre-remediation baseline data also prevented before-and-after analysis at the most upstream site, Plymouth (U). Because the remediation efforts evaluated in this study were completed only one season before monitoring was expanded, some responses may not yet have been fully expressed as observed in other restoration studies (Siemion and others, 2016; Collins and others, 2013). Continued monitoring could improve the ability to evaluate longer-term effects of remediation on water quality in the Yellow River.

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Appendix 1. Statistical Output Summaries

Appendix 1.1. Statistical summary of generalized least square regression (GLS) models comparing turbidity-discharge relations (test 1a) and mean discharge (test 1b), as well as results from a linear mixed effects (LME) model comparing suspended sediment concentration, between periods before and after remediation in the Yellow River, Indiana. Bolded p-values denote fixed effects of interest and asterisks denote fixed effects of interest that were significant ($p < 0.05$).

[FNU, Formazin Nephelometric Units, Test described in table 5. Data used in analysis available at U.S. Geological Survey 2025.]

Test	Response Variable	Model Parameter	Oak Grove			Knox		
			Value	Standard Error	p-value	Value	Standard Error	p-value
Test 1a	Mean daily turbidity, in FNU	Intercept	-6.76	0.24	0.00	-9.16	0.33	0.00
		Discharge	1.56	0.04	0.09	1.92	0.05	0.00
		Period	0.59	0.35	0.00	2.19	0.56	0.00
		Period x Discharge	-0.12	0.06	0.05*	-0.38	0.08	0.00*
Test 1b	Mean daily discharge, in cubic feet per seconds	Intercept	5.64	0.03	0.00	5.73	0.01	0.00
		Period	-0.17	0.08	0.04*	0.15	0.12	0.18
Test 1c	Suspended sediment concentration, in milligrams per liter	Intercept	-1.48	0.59	0.01	-3.03	0.58	0.00
		Discharge	0.81	0.10	0.00	1.09	0.09	0.00
		Period	0.09	0.21	0.67	0.00	0.20	0.99

Appendix 1.2. Statistical summary of generalized least square regression (GLS) models comparing turbidity-discharge relations (test 2a) and mean discharge (test 2b) among sites in post-remediation period. Bolded p-values denote fixed effects of interest and asterisks denote fixed effects of interest that were significant ($p < 0.05$).

[FNU, Formazin Nephelometric Units; <, less than; Test described in table 6.
Data used in analysis available at U.S. Geological Survey 2025.]

Test	Model Response Variable	Model Parameter	Value	Standard Error	p-value
Test 2a ^a	Mean daily turbidity, in FNU	Intercept	-5.17	0.51	<0.001
		Discharge	1.38	0.09	<0.001
		Site:Knox	-2.61	0.77	<0.001*
		Site:Oak Grove	-2.45	0.74	<0.001*
		Discharge x Knox	0.28	0.12	0.02*
		Discharge x Oak Grove	0.27	0.12	0.02*
Test 2b	Mean daily discharge, in cubic feet per second	Intercept	5.23	0.55	<0.001
		Site:Knox	0.67	0.77	0.38
		Site:Oak Grove	0.44	0.77	0.57

^a Coefficients for site represent differences in intercept relative to the reference site (Plymouth), while the interaction terms represent differences in the turbidity-discharge slope relative to the reference site (Plymouth).

Appendix 1.3. Site fixed-effect estimates from linear mixed effects regression models comparing mean water-quality concentrations among five sites on the Yellow River in the post-remediation period (test 2c). For each parameter, models were fit with site as a fixed effect, natural log-transformed discharge as a covariate, and with random effects specified as described in the Methods. Reported coefficient estimates represent differences in mean natural log-transformed concentration at each site relative to the reference site (Plymouth), adjusted for discharge and date-level random effects. Pairwise comparisons among site estimated marginal means (Tukey-adjusted) are presented in table 9 of the report.

[N, nitrogen; P, phosphorus; Rd, Road, CR1100E; County Rd 1100 East; <, less than. Site short name from table 1. Data used in analysis from U.S. Geological Survey, 2025.]

Response Variable	Site	Estimate	Standard Error	p-value
Suspended sediment concentration, in milligrams per liter	Upas Rd	-0.30	0.13	0.02
	Oak Grove	-0.43	0.13	< 0.01
	CR1100E	-0.30	0.13	0.02
	Knox	-0.32	0.15	0.03
Total Nitrogen as N, in milligrams per liter	Upas Rd	-0.13	0.03	< 0.01
	Oak Grove	-0.16	0.03	< 0.01
	CR1100E	-0.15	0.03	< 0.01
	Knox	-0.40	0.04	< 0.01
Nitrate + nitrite as N, in milligrams per liter	Upas Rd	-0.11	0.03	< 0.01
	Oak Grove	-0.13	0.03	< 0.01
	CR1100E	-0.11	0.03	< 0.01
	Knox	-0.39	0.04	< 0.01
Nitrite as N, in milligrams per liter	Upas Rd	-0.27	0.08	< 0.01
	Oak Grove	-0.34	0.08	< 0.01
	CR1100E	-0.32	0.08	< 0.01
	Knox	-0.46	0.10	< 0.01
Total ammonia as N, in milligrams per liter	Upas Rd	-0.38	0.08	< 0.01
	Oak Grove	-0.44	0.09	< 0.01
	CR1100E	-0.51	0.09	< 0.01
	Knox	-0.62	0.10	< 0.01
Total Phosphorus as P, in milligrams per liter	Upas Rd	-0.21	0.06	< 0.01
	Oak Grove	-0.27	0.07	< 0.01
	CR1100E	-0.29	0.06	< 0.01
	Knox	-0.49	0.08	< 0.01
Orthophosphate as P, in milligrams per liter	Upas Rd	-0.14	0.08	0.10
	Oak Grove	-0.17	0.09	0.05
	CR1100E	-0.22	0.09	0.01
	Knox	-0.50	0.10	< 0.01

^a Site coefficients are relative to the reference site (Plymouth).

Negative estimates indicate lower concentrations than the reference site; positive estimates indicate higher concentrations.

^b Responses were natural log-transformed prior to modeling. Coefficients are presented on the log scale.

^c All models included natural log-transformed discharge as a covariate; discharge coefficients are not shown for brevity.

^d A random intercept for sampling date was included to account for non-independence of repeated measurements through time.

^e Likelihood ratio tests comparing models with and without the site term were conducted using maximum likelihood fits; fixed-effects estimates shown here are from restricted maximum likelihood fits.

Appendix 1.4. Site fixed-effect estimates from linear mixed effects regression models comparing mean water-quality fluxes among five sites on the Yellow River in the post-remediation period (test 2c). For each parameter, models were fit with site as a fixed effect, season as a fixed effect to account for flow variability, and with random effects specified as described in the Methods. Reported coefficient estimates represent differences in mean natural log-transformed flux at each site relative to the reference site (Plymouth), adjusted for season and date-level random effects. Pairwise comparisons among site estimated marginal means (Tukey-adjusted) are presented in table 10 of the report.

[N, nitrogen; P, phosphorus; Rd, Road, CR1100E; County Rd 1100 East; <, less than. Site short names from table 1. Data used in analysis from U.S. Geological Survey, 2025.]

Response Variable	Site	Estimate	Standard Error	p-value
Suspended sediment concentration, in kilograms per hour	Upas Rd	0.33	0.14	0.02
	Oak Grove	0.32	0.14	0.03
	CR1100E	0.36	0.15	0.02
	Knox	0.79	0.15	< 0.01
Total Nitrogen as N, in kilograms per hour	Upas Rd	0.34	0.08	< 0.01
	Oak Grove	0.40	0.08	< 0.01
	CR1100E	0.35	0.08	< 0.01
	Knox	0.44	0.08	< 0.01
Nitrate + nitrite as N, in kilograms per hour	Upas Rd	0.36	0.08	< 0.01
	Oak Grove	0.43	0.08	< 0.01
	CR1100E	0.37	0.08	< 0.01
	Knox	0.44	0.08	< 0.01
Nitrite as N, in kilograms per hour	Upas Rd	0.09	0.07	0.24
	Oak Grove	0.08	0.07	0.30
	CR1100E	0.05	0.08	0.50
	Knox	0.16	0.08	0.03
Total ammonia as N, in kilograms per hour	Upas Rd	0.05	0.09	0.55
	Oak Grove	0.08	0.09	0.37
	CR1100E	-0.05	0.09	0.54
	Knox	0.15	0.09	0.10
Total Phosphorus as P, in kilograms per hour	Upas Rd	0.34	0.09	< 0.01
	Oak Grove	0.38	0.09	< 0.01
	CR1100E	0.29	0.09	< 0.01
	Knox	0.49	0.09	< 0.01
Orthophosphate as P, in kilograms per hour	Upas Rd	0.39	0.11	< 0.01
	Oak Grove	0.46	0.11	< 0.01
	CR1100E	0.33	0.11	< 0.01
	Knox	0.43	0.11	< 0.01

a Site coefficients are relative to the reference site (Plymouth). Negative estimates indicate lower flux than the reference site; positive estimates indicate higher flux.

b Responses were natural log-transformed prior to modeling. Coefficients are presented on the log scale.

c All models included season as a fixed effect to account for flow variability; season coefficients are not shown for brevity.

d A random intercept for sampling date was included to account for non-independence of repeated measurements through time.

e Likelihood ratio tests comparing models with and without the site term were conducted using maximum likelihood fits; fixed-effects estimates shown here are from restricted maximum likelihood fits.

Appendix 1.5. Summary of statistical results from log-likelihood ratio tests between a full linear mixed effects (LME) model where site order was included as a fixed effect and a null LME model where site order was not included for sediment and nutrient concentrations and loads in the Yellow River remediation study in northwest Indiana.

[N, nitrogen; P, phosphorus; < less than. Data used in analysis available at U.S. Geological Survey, 2025]

Constituent	Chi-square	p-value
Discharge, in cubic feet per second		
Discharge	26.6	<0.01
Concentration, in milligrams per liter		
Suspended Sediment	2.82	0.093
Total Nitrogen as N	44.0	<0.01
Nitrate + Nitrite as N	43.0	<0.01
Nitrite as N	12.7	<0.01
Total Ammonia as N	22.8	<0.01
Total Phosphorus as P	26.8	<0.01
Orthophosphate as P	21.3	<0.01
Flux, in kilograms per hour		
Suspended Sediment	20.7	<0.01
Total Nitrogen as N	20.5	<0.01
Nitrate + Nitrite as N	20.5	<0.01
Nitrite as N	2.93	0.09
Total Ammonia as N	0.899	0.34
Total Phosphorus as P	15.9	<0.01
Orthophosphate as P	8.80	<0.01